

CONCEPT NOTE

Cyber-Physical Solar Energy Platform: From Bench Prototype to Microgrid Resilient Monitoring Node

Author: Jonny da Conceição Giaretta

Field: Computer & Electrical Engineering (Cyber-Physical Systems & Energy Infrastructure)

Target Research Alignment: Dr. Preetham Goli (Power System Monitoring, Microgrids & Cyber-Physical Security)

Document Type: Two-Page Concept Note for Academic Guidance & Research Exploration

1. Executive Summary & Research Motivation

Modern distribution networks and microgrids are rapidly shifting toward decentralized, inverter-dominated architectures. While distributed energy resources (DERs) offer unprecedented sustainability, they introduce complex operational dynamics: volatile generation profiles, bi-directional power flows, and heightened vulnerability to cyber-physical disturbances.

Core Motivation & Purpose:

This project was conceived not merely as a standalone hardware exercise, but as a modular, low-cost **Cyber-Physical Testbed** for energy-aware edge control, real-time telemetry, and microgrid power monitoring. My primary motivation is to bridge physical power hardware (photovoltaic generation, battery management, and power electronics) with embedded logic, firmware design, and software architecture.

Rather than treating renewable hardware as static devices, this platform explores how we can build resilient, self-monitoring edge nodes capable of supporting microgrid stability, local fault detection, and cybersecurity in modern power networks.

2. System Architecture & Evolutionary Roadmap

The platform follows a four-phase engineering approach to ensure physical safety, deterministic validation, and systemic scalability.

Phase	Focus Domain	Key Hardware & Firmware Implementation	Current Status
Phase 1	Bench Harvesting	6 V/5 W	Completed

	& Safety MVP	Photovoltaic panel, TP4056 charge controller, 3.7 V nominal Li-ion cell (18650), high-frequency DC-DC boost converter ($5.0\text{ V} \pm$ output), and isolation protocols.	
Phase 2	Edge Automation & Tracking	Quad LDR differential sensing array, dual-axis servo tracking (15° — horizontal, 20° — vertical), C++ control loop (10 Hz sampling), and serial CSV telemetry pipeline.	Active Prototype
Phase 3	Scaled Power Bank & Storage	$30\text{ W}/18\text{ V}_{\text{PV}}$ input, 3S2P Li-ion bank (11.1 V nominal / 12.6 V peak), 3S 40 A active-balancing BMS, and synchronous Buck regulators for compute power.	In Progress
Phase 4	Cyber-Physical & Telemetry	MQTT/LoRaWAN communication	Target Horizon

		pipeline, time-series data storage, on-node anomaly detection, and evaluation of sensor tampering/FDIA resilience.	
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3. Key Engineering Decisions & Bench Methodology

1. **Overhead Voltage Buffer:** Utilizing a 6 V nominal panel for a 4.2 V Li-ion cutoff provides the necessary diode and switching overhead required under partial cloud coverage and variable irradiance.
2. **Switching vs. Linear Regulation:** Eliminating linear regulators (e.g., LM7805) in favor of magnetic inductor-based switching Boost/Buck converters maximizes thermodynamic efficiency and minimizes thermal dissipation.
3. **Decoupled Power Rails:** Decoupling microcontroller logic power (5 V regulated) from motor actuator supply rails prevents inductive back-EMF spikes and brownout resets during transient motor startup currents.

4. Potential Research Alignment with Dr. Goli's Laboratory

Given Dr. Goli's leadership in distribution system operation, microgrid protection, power monitoring, and cyber-physical security, I envision this physical platform as an open experimental foundation to explore several academic vectors:

- **Vector A: Edge Power Quality & State Estimation Monitoring:**
Adapting embedded telemetry to monitor localized voltage harmonics, state-of-charge (SoC) degradation, and dynamic solar irradiance fluctuations, serving as a low-cost, physical micro-PMU (Phasor Measurement Unit) concept node.
- **Vector B: Cyber-Physical Resilience & Anomaly Detection:**
Simulating physical sensor tampering or telemetry corruption at the edge to test adaptive filtering algorithms and evaluate system resilience in islanded microgrid scenarios.
- **Vector C: Hardware-in-the-Loop (HIL) Co-Simulation Integration:**
Interfacing the hardware prototype's real-time generation feed with open-source distribution simulation tools (such as **OpenDSS**) to model how physical solar dynamics impact low-voltage distribution feeders.

5. Learning & Academic Exploration Goals

As an undergraduate student eager to ground my theoretical learning in real-world engineering challenges, my primary goals in sharing this document are to:

1. Receive Dr. Goli's technical critique on the power management architecture and sensor telemetry pipeline.
2. Learn how bench-scale hardware prototypes can be rigorously aligned with microgrid protection and co-simulation standards.

Author & Contact Information

Jonny da Conceição Giaretta

Computer & Electrical Engineering Student

- **Email:** jonny.da.conceicao.giaretta@gmail.com
- **GitHub:** github.com/jonnygiaretta
- **LinkedIn:** linkedin.com/in/jonny-da-conceicao-giaretta
- **YouTube:** [youtube.com/@Jonny Giaretta](https://youtube.com/@Jonny_Giaretta) (Project demonstrations, technical build logs, and video updates will be hosted here)